# Alexandros Sigaras

Director, AI/XR,
Englander Institute for Precision Medicine
Assistant Professor of Research in
Systems and Computational Biomedicine

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## Research Interests

My scientific focus is on building secure and AI driven high scalable computing solutions in the cloud for multimodal medical “Big Data” specializing in genomic analysis for cancer care and cancer research. Furthermore, to visualize these complex big data entities, I am also interested in spatial computing, Mixed Reality, Virtual Reality and Augmented Reality as solutions to enable immersive asynchronous remote collaboration and data visualization of healthcare and genomic data.

## Education

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| December 2013 | **M.S. in Computer Science**, *Columbia University*, Department of Computer Science, Fu Foundation School of Engineering and Applied Science, New York, NY, USA.<br>**Track:** Thesis Research Track<br>**GPA:** 3.851/4<br>**Thesis:** *Surgical Structured Light for 3D Minimally Invasive Surgical Imaging* |
| October 2011 | **B.S. in Informatics,** *University of Piraeus,* Department of Informatics, Piraeus, Greece.<br>**Track:** Information Systems Track<br>**GPA:** 8.5/10<br>**Thesis:** *Self Adaptive Robotic Warehouse Management Systems with event and location-based triggers* |

## Experience

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| Weill Cornell Medical College | **Assistant Professor of Research in Systems and Computational Biomedicine,**<br>**Department of Systems and Computational Biomedicine** |
| August 2025 - Present | **Director, AI/XR Lab**<br>**Englander Institute for Precision Medicine** |

- Maintain, develop and manage efforts pertaining to the Laboratory for Artificial Intelligence and Extended Reality (AI/XR Lab) including leading the development of novel software applications in precision medicine.

- Lead the design and development of novel software solutions supporting the clinical and research activities at the Joint Clinical Genomics Initiative and at the Englander Institute for Precision Medicine.
- Oversee the design, development, and deployment of containerized solutions for the clinical genomics laboratory management system and data analysis, including QC/QA tools.
- Lead the efforts of large-scale genomic analysis in the cloud for both clinic and research activities; coordinate the development of artificial intelligence solutions to support the precision medicine knowledge base and the case review and sign out process.

Weill Cornell Medical College

**Director, Artificial Intelligence and Extended Reality (AI/XR)**
**Englander Institute for Precision Medicine**

April 2023 - Present

- Lead AI/XR research at the institution, establishing a research hub for AI/XR at the EIPM, as well as pursue funding opportunities for the institute's research initiatives.
- Develop and implement AI/XR strategies that align with the institution's overall goals and objectives.
- Manage the development and implementation of AI/XR projects, including managing budgets, timelines, and resources.
- Collaborate with clinicians, researchers, and students across Cornell campuses to identify opportunities for AI/XR to improve patient care, clinical workflows, and research outcomes, as well as develop a new syllabus to teach the future of medicine.
- Ensure the ethical use of AI applications, protecting patient privacy and avoiding biases that could lead to discrimination or harm.
- Stay ahead of the curve with AI/XR developments and research by attending conferences and staying abreast of emerging trends and technologies.
- Engage with external partners, such as industry partners and other academic institutions, to foster collaboration and identify opportunities for joint research and development projects.

Weill Cornell Medical College

April 2020 - Present

**Assistant Professor of Research in Physiology and Biophysics,**
**Assistant Professor of Research in Computational Biomedicine,**
**Department of Physiology and Biophysics**
**Director, AI/XR Lab**
**Institute for Computational Biomedicine,**
**Englander Institute for Precision Medicine**

- Maintain, develop and manage efforts pertaining to the Laboratory for Artificial Intelligence and Extended Reality (AI/XR Lab) including leading the development of novel software applications in precision medicine.
- Lead the design and development of novel software solutions supporting the clinical and research activities at the Joint Clinical Genomics Initiative and at the Englander Institute for Precision Medicine.
- Oversee the design, development, and deployment of containerized solutions for the clinical genomics laboratory management system and data analysis, including QC/QA tools.

- Lead the efforts of large-scale genomic analysis in the cloud for both clinic and research activities; coordinate the development of artificial intelligence solutions to support the precision medicine knowledge base and the case review and sign out process.

Weill Cornell Medical College

May 2017 – April 2020

**Senior Research Associate in Computational Biomedicine,**
**Department of Physiology and Biophysics**
**Institute for Computational Biomedicine,**
**Englander Institute for Precision Medicine**

- Lead the design and development of novel software solutions supporting the clinical and research activities at the Joint Clinical Genomics Initiative and at the Englander Institute for Precision Medicine.
- Lead the development of the virtual and augmented reality program for precision medicine.
- Oversee the design, development, and deployment of containerized solutions for the clinical genomics laboratory management system and data analysis, including QC/QA tools
- Lead the efforts of large-scale genomic analysis in the cloud for both clinic and research activities; coordinate the development of artificial intelligence solutions to support the precision medicine knowledge base and the case review and sign out process.

Weill Cornell Medical College

April 2015 – May 2017

**Research Associate in Computational Biomedicine,**
**Department of Physiology and Biophysics**
**Institute for Computational Biomedicine,**
**Englander Institute for Precision Medicine**

- Lead the search for new ways to leverage technology to advance precision medicine, including the use of Virtual Reality and Augmented Reality.
- Responsible for the design and implementation of the informatics infrastructure of the Englander Institute for Precision Medicine (EIPM), including the design and development of new tools to simplify user interaction with the laboratory information management system and the current IT infrastructure of WCM/NYP, collect and analyze user data and system performance within EIPM, and implement tools to report the results of the genomic analysis to the users via genomic portals.

Weill Cornell Medical College

March 2014 – April 2015

**Programmer Analyst,**
**Department of Pathology**
**Institute for Computational Biomedicine,**
**Englander Institute for Precision Medicine**

Software Development Lead for all clinical and R&D activities of the Englander institution for Precision Medicine.
**Key Responsibilities:** Development of Next-Generation sequencing pipelines, LIS, cancer genomic portals and integrating genomic data to New York Presbyterian's HL7 infrastructure.

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| Columbia University | **Research Assistant (graduate), Robotics Lab** |
| September 2012 – March 2014 | ▪ **Project 1:** Developed prototype on Surgical Structured Light for 3D minimally invasive surgical imaging.<br>▪ **Project 2:** Developed brain computer interfaces for robotic grasping for people with locked-in syndrome. |
| Microsoft | **Internal Sales Education Specialist** |
| December 2011 – July 2012 | Worked closely with universities and K-12 schools in Greece to provide cloud solutions and generally fulfill education institution needs in IT. Recipient of the Language Champion Award for the contribution to the Windows 8 International Review Program. |
| Microsoft | **Microsoft Student Partner** |
| November 2006 – December 2011 | Member of Developers Platform Evangelists (DPE) group. Tasks included: administering the departmental Microsoft Developer Network Academic Alliance (MSDNAA) subscription, organizing technical presentations (for students) involving Microsoft products, advising students entering Microsoft's worldwide "Imagine Cup" programming contest, and setting up and moderating the studentguru.gr community website. |

# Support

## Current Funding

**NIH 1OT2OD032720-01** *"Bridge2AI: Voice as a Biomarker of Health - Building an ethically sourced, bioaccoustic database to understand disease like never before"* **Sigaras Alexandros (Co-I and Module Lead)**.
Duration: 09/2022 – 11/2026.

- Total costs for 2022 – 2023: $3,817,302.
- Total costs for 2023 – 2024: $5,302,032.
- Total costs for 2024 – 2025: $4,215,002.
- Total costs for 2025 – 2026: $4,660,942.

# Honors and Awards

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| November 2015 | **Nominee**, Forbes 30 under 30 in the Science and Healthcare categories |
| July 2012 | **Language Champion Award** for Contribution to the Windows 8 International Review Program |
| June 2012 | **Fulbright Scholar,** U.S Department of State's Bureau of Educational and Cultural Affairs |

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| June 2012 | **Scholarship** (for M.S. studies), Harry D. Triantafillu Scholarship Fund Award, Institute of International Education |
| October 2011 | **Salutatorian**, University of Piraeus, Department of Informatics |
| October 2011 | Graduated **summa cum laude**, University of Piraeus, Department of Informatics |
| November 2010 | **Semi-Finalist**, Imagine Cup IT Challenge, Microsoft |
| December 2009 | **Honorary Scholarship**, Hellenic American University |
| December 2009 | **2nd place winner**, Athens Startup Weekend 2 |
| November 2008 | **Semi-Finalist**, Imagine Cup IT Challenge, Microsoft |

# Teaching

## Teaching Assistant

- Created and graded homework, midterms, and final exams; prepared project material and gave lectures (Numbers in parentheses indicate enrollment).

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| COMS W4733 | Computational Aspects of Robotics, Columbia University<br>Instructor: Prof. Peter Allen, Fall 2013 (60) |

# Certifications

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| December 2011 | Udacity – Introduction to Artificial Intelligence - (top 10%) |
| October 2010 | MCP, MCTS - 70-680: Microsoft Technology Specialist – Windows 7, Configuration |
| June 2007 | MCP, MCTS - 70-680: Microsoft Technology Specialist – Windows Vista, Configuration |

# Professional Service and Academic Leadership

## Scientific Review Boards and Committees

- Member, Scientific Review Committee (SRC), Weill Cornell Medicine-Qatar (WCM-Q) , Doha, Qatar May 2025 – Present
    - Review Editor for “Virtual Reality in Medicine” – Journal of Frontiers in Virtual Reality Guest Editor for “XR in Surgical and Procedural Medicine” – Journal of Medical Extended Reality

## Editorial and Peer Review Activities

- Review Editor for “Virtual Reality in Medicine” – Journal of Frontiers in Virtual Reality Guest Editor for “XR in Surgical and Procedural Medicine” – Journal of Medical Extended Reality
- Guest Editor Assistant for “Clinical Medicine and Surgery in the Digital Age: Advances in Virtual, Augmented, Mixed, and Extended Reality” – Journal of Clinical Medicine
- Reviewer – Nature

# Publications

## Journal Articles

[J.56] Bensoussan Y, **Sigaras A**, Rameau A, Elemento O, Powell M, Dorr D, Payne P, Ravitsky V, Bélisle-Pipon JC, Bahr R, Watts S, Bolser D, Siu J, Lerner-Ellis J, Rudzicz F, Boyer M, Abdel-Aty Y, Syed TA, Anibal J, Aradi S, Armosh K, Martinez AS, Awan S, Bedrick S, Beltran H, Bernier A, Berrios M, Bevers I, Blatter A, Brito R, Brown A, Brown J, Casalino S, Costello J, Dalal A, De Santiago I, Diaz-Ocampo E, Doerksen E, Ebraheem M, Eiseman E, Elmahdy M, Evangelista E, Fletcher K, Gallois H, Garrett G, Gelbard A, Ghaffar O, Ghavanini A, Goldenberg A, Hanna K, Hersh W, Jain J, Jayachandran L, Jenney K, Jenkins K, Jenkins P, Jo S, Johnson A, Kalia A, Kalia M, Khawa Z, Kobayashi K, Kostelnik C, Krause A, Krussel A, Lapadula E, Leo G, Levinsky J, Loewith C, Mah L, Maharaj V, Maramraju S, Miao S, Michaels L, Mifsud M, Mikhael M, Moothedan E, Nafii Y, Neal T, Newberry K, Ng E, Nickel C, Peltier A, Persaud J, Pharr T, Pontell M, Potter J, Premi-Bortolotto C, Rafatjou P, Rahman JM, Rajkumar G, Rajkumari S, Ramos J, de Riesthal M, Rohde S, Rossi J, Russell L, Salvi Cruz S, Samuel J, Shah S, Shawkat A, Silberholz E, Stark J, Su L, Sudhakar SG, Sutherland D, Mukhi VS, Tang J, Taylor L, Toghranegar J, Tu J, Urbano M, Vinson K, Wilke J, Wilson C, Zanin M, Zeng X, Zesiewicz T, Zhao R, Zisimopoulos P, Ghosh S. Bridge2AI-Voice: An ethically-sourced, diverse voice dataset linked to health information. ***PhysioNet***. 2026 Oct 19; version 4.0.0. doi: 10.13026/37yb-1t42.

[J.55] Sudhakar SG, Musteric S, Watts S, Ebraheem M, Templeton JM; **Bridge2AI-Voice Consortium**; Bensoussan Y. A Comparative Study of Acoustic Feature Extraction Using CSL and Web Prototype of LIS-N Application. ***medRxiv***. 2026 Sep 14:2026.09.14.26362731. doi: 10.64898/2026.09.14.26362731. PMID: 42820718.

[J.54] Anibal J, Nama GKC, Ganesh S, Nafii Y, Salvi Cruz S, Daoud V, Zanin M, Le T, Khorrami P, Wood BJ, Clifton D, Toghranegar J, Bensoussan Y; **Bridge2AI-Voice Consortium**. Large Language Models for Optimizing Patient Recruitment Decisions in Voice Data Generation Projects. ***Laryngoscope Investig Otolaryngol***. 2026 Aug 24;11(4):e70519. doi: 10.1002/lio2.70519. PMID: 42639494; PMCID: PMC13502059.

[J.53] Annabestani M, Sriram S, Kuzemczak A, Wong SC, **Sigaras A**, Mosadegh B. Enhanced Real-Time 6-DOF Extended Reality Catheter Tracking for Evaluating Potential Improvement in Efficiency, Precision, and Depth Perception for Cardiac Interventions. ***arXiv preprint*** arXiv:2608.07606. 2026 Aug.

[J.52] Jotwani R, Barot VD, Goldstein PA, **Sigaras A**, Sriram S, White RS, Mukherjee D, Gabbay E, Chan JM, Steigerwald Schnall J, Rubin JE. A Conversational Ethics Bot (CALEB) Versus GPT-4o for Medical Ethics Education: Transcript Analysis and Extended Reality Feasibility Study. ***Journal of Medical Extended Reality***. 2026 Jul 16;3:29941520261469342. doi: 10.1177/29941520261469342. PMID: 42558094.

[J.51] Jenkins PD, Bedrick S, Karstens L, Hersh W; **Bridge2AI-Voice Consortium**; Dorr DA. From voice biomarkers to telemedicine screening: developing and evaluating a voice-based AI model for laryngeal lesion detection using the Bridge2AI-Voice dataset. ***Front Digit Health***. 2026 Jul 1;8:1846369. doi: 10.3389/fdgth.2026.1846369. PMID: 42459998.

[J.50] Kutler RB, Gantwerker E, Jotwani R, **Sigaras A**, Difede J, Rameau A. Extended Reality in Otolaryngology–Head & Neck Surgery: A State-of-the-Art Review. ***Otolaryngol Head Neck Surg***. 2026 Oct;175(4):802-824. doi: 10.1002/ohn.70323. PMID: 42371620.

[J.49] Ebraheem M, Toghranegar J; **Bridge2AI-Voice Consortium**; Bensoussan Y, Templeton JM. Explainable and Interpretable AI for Voice and Speech Analysis in Clinical Care: Systematic Review. ***J Med Internet Res***. 2026 Jun 24;28:e83790. doi: 10.2196/83790. PMID: 42341346.

[J.48] Elemento O, **Sigaras A**, Colonel J, Hajirasouliha I, Ghosh S, Bensoussan Y; **Bridge2AI-Voice Consortium**; Rameau A. Towards A Foundation Model for Clinical Voice Biomarkers. ***medRxiv***. 2026 May 28:2026.05.28.26354346. doi: 10.64898/2026.05.28.26354346.

[J.47] Waltman P, Chandra P, Eng KW, Wilkes DC, Park H, Pabon C, Delpe P, Bhinder B, Manohar J, Kane T, Fernandez E, Gorski K, Greco N, Simi M, Tang JM, Zisimopoulos P, King A, Al Assaad M, Ten Eyck T, Roberts D, Monge J, Demichelis F, Tam W, Ouseph MM, **Sigaras A**, Beltran H, Rennert H, Lindeman N, Song W, Solomon J, Mosquera JM, Kim R, Catalano J, Hassane DC, Sigouros M, Elemento O, Alonso A, Sboner A. EXaCT-2: an augmented and customizable oncology-focused whole exome sequencing platform. ***NPJ Precis Oncol***. 2026 Apr 21. doi: 10.1038/s41698-026-01390-5. PMID: 42014463.

[J.46] Kozan EN, Delgado-de la Mora J, Al Assaad M, Ginter P, Gundem G, Levine MF, Gorski K, Phan A, Beg S, **Sigaras A**, LaPolla P, Semaan A, Wilkes DC, Greco N, Sigouros M, Manohar J, Medina-Martínez JS, Sboner A, Elemento O, Hoda S, Saxena A, Andreopoulou E, Boyraz B, Mosquera JM. Whole genome sequencing of locally advanced and metastatic breast carcinoma unravels relevant molecular signatures and novel events. ***Pathol Res Pract***. 2026 Jun;282:156469. doi: 10.1016/j.prp.2026.156469. PMID: 42013747.

[J.45] Henry O, Brumberger E, Jotwani R, **Sigaras A**, Sriram S, Yeh IM, Rubin J. Using an Artificial Intelligence Conversational Agent in Virtual Reality to Teach Serious Illness Communication Skills to Residents. ***J Grad Med Educ***. 2026 Apr;18(2):193-194. doi: 10.4300/JGME-D-25-00952.1. PMID: 42005895; PMCID: PMC13086130.

[J.44] Salvi Cruz S, Toghranegar J, Malin B, Mehra T, MacDonald B, Speights M, Noufi C, Fossat Y, Fagherrazi G, Cummins N, Elbeji A, Blatter A, Gelbard A, Arienzo A, Baur S, Wetstone K, Peller J, Au R, Botha H, Lahav A, Hemmerling D, Catania F, Anibal J, Saha S, Coleman O, Gallois H, Avila Martinez S, Mahapatra N, Sara JDS, Mathur A, Patel R, Zieliński K, Kourtis L, Lerner-Ellis J, Cong Y, Ngo H, Talkar T, Hale G, Comito K, Ghosh S, Watts S, Bedrick S, Powell M, Bélisle-Pipon JC, Krussel A, Mahapatra I, Bahr R, Hanna K, Kostelnik C, Dorsey K, John M, Curp K, Rameau A; **Bridge2AI-Voice Consortium**; Bensoussan Y. Translating AI research into reality: summary of the 2025 voice AI Symposium and Hackathon. ***Front Digit Health***. 2026 Mar 16;8:1754426. doi: 10.3389/fdgth.2026.1754426. PMID: 41918982.

[J.43] Malo MF, Bouhouita-Guermech S, Gallois H, Ravitsky V; **Bridge2AI-Voice Consortium**; Bélisle-Pipon JC. Using voice and speech data in healthcare: a scoping review of the ethical, legal and social implications. ***Front Digit Health***. 2026 Feb 10;8:1750111. doi: 10.3389/fdgth.2026.1750111. PMID: 41743677; PMCID: PMC12930345.

[J.42] He L, Li H, Wang S, Baik E, Kervin S, Zhao R, Ramos JM; **Bridge2AI-Voice Consortium**; Colonel J, Rameau A. Decoding Gender in Cough Sounds: A Transformer-Based Analysis. ***Laryngoscope***. 2026 Jan 26. doi: 10.1002/lary.70393. Epub ahead of print. PMID: 41588706.

[J.41] Gallois H, Ivkovic L, Evangelista E, Bensoussan Y; **Bridge2AI-Voice Consortium**; Dorr DA, Elemento O, Ghosh SS, Johnson A, Payne P, Powell ME, Rameau A, Ravitsky V, **Sigaras A**, Bélisle-Pipon JC. "Low Risk, High Happiness": A Review of Openly Declared Ethical and Legal Practices in Voice Biomarker Health-Tech Start-Ups. ***Health Care Anal***. 2026 Sep;34(3):744-782. doi: 10.1007/s10728-025-00539-w. PMID: 41091343.

[J.40] Dorr DA, Krussel A, Hauck R, Jackson C, Dalal A, Bedrick S, Payne PRO; **Bridge2AI-Voice Consortium**; Hersh W. Adapting data science competencies by role and purpose: Voice AI. ***Front Digit Health***. 2025 Oct 21;7:1610253. doi: 10.3389/fdgth.2025.1610253. PMID: 41195285; PMCID: PMC12582912.

[J.39] Rizzo A, Mozgai S, **Sigaras A**, Rubin JE, Jotwani R. Expert Consensus Best Practices for the Safe, Ethical, and Effective Design and Implementation of Artificially Intelligent Conversational Agent (i.e. Chatbot/Virtual Human) Systems in Healthcare Applications. ***Journal of Medical Extended Reality*** 2025 Sep 8 (Vol. 2, No. 1, pp. 209-222) doi: 10.1177/29941520251369450

[J.38] Jenkins P, Harrison R, Bedrick S, Karstens L; **Bridge2AI-Voice Consortium**; Hersh W. Voice as a biomarker: exploratory analysis for benign and malignant vocal fold lesions. ***Front Digit Health.*** 2025 Aug 12;7:1609811. doi: 10.3389/fdgth.2025.1609811. PMID: 40873766; PMCID: PMC12378753.

[J.37] Awan SN, Bensoussan Y, Watts S, Boyer M, Budinsky R, Bahr RH; **Bridge2AI-Voice Consortium**. Influence of recording instrumentation on measurements of voice in sentence contexts: use of smartphones and tablets. ***Front Digit Health***. 2025 Jul 9;7:1610772. doi: 10.3389/fdgth.2025.1610772. PMID: 40704366.

[J.36] Blatter A, Gallois H, Evangelista E, Bensoussan Y; **Bridge2AI-Voice Consortium**; Bélisle-Pipon JC. "Voice is the New Blood": a discourse analysis of voice AI health-tech start-up websites. ***Front Digit Health.*** 2025 May 29;7:1568159. doi: 10.3389/fdgth.2025.1568159. PMID: 40510414; PMCID: PMC12158947.

[J.35] Anibal J, Doctor R, Boyer M, Newberry K, De Santiago I, Awan S, Abdel-Aty Y, Dion G, Daoud V, Huth H, Watts S, Wood BJ, Clifton D, Gelbard A, Powell M, Toghranegar J; **Bridge2AI Voice Consortium**; Bensoussan Y. Transformers for rapid detection of airway stenosis and stridor. ***Sci Rep***. 2025 May 2;15(1):15394. doi: 10.1038/s41598-025-99369-y. PMID: 40316652; PMCID: PMC12048665.

[J.34] Moothedan E, Boyer M, Watts S, Abdel-Aty Y, Ghosh S, Rameau A, **Sigaras A**, Elemento O; Bridge2AI-Voice Consortium; Bensoussan Y. The Bridge2AI-voice application: initial feasibility study of voice data acquisition through mobile health. ***Front Digit Health.*** 2025 Apr 15;7:1514971. doi: 10.3389/fdgth.2025.1514971. PMID: 40302934; PMCID: PMC12037532.

[J.33] Bélisle-Pipon JC, Anibal J, Bahr R, Bedrick S, Coleman O, Dorr D, Evans BJ, Fagherazzi G, Gelbard A, Ghosh S, Ho A, Jackson C, Joachim D, Kourtis L, Krussel A, Lahav A, Leuze B, MacDonald B, Miller G, Mohan V, Naunheim M, Powell M, Rameau A, Ramphal S, Ravitsky V, Reavis C, Salvi Cruz S, Toghranegar J, Vogel A, Watts S, Yracheta J, Zhao R; **Bridge2AI-Voice Consortium**; Bensoussan Y. Interactive Panel Summaries of the 2024 Voice AI Symposium. ***Front Digit Health***. 2025 Mar 27;7:1484521. doi: 10.3389/fdgth.2025.1484521. PMID: 40212896; PMCID: PMC11983451.

[J.32] Bhavsar N, Sriram S, Balasubramanian S, Davidson C, Piechowski W, Zimm J, Sanapala S, Fortenko A, Lame M, Greenwald PW, St. George J, **Sigaras A.** AIRWAY-XR: Augmented Instruction to Refine Wayfinding and Yielding Skills in Emergency Medicine Residents for Intubation using Mixed Reality Technology. ***medRxiv***. 2025 Jan 6:2025-01.

[J.31] Waltman P, Chandra P, Eng KW, Wilkes DC, Park H, Pabon C, Gorski K, Greco N, SimiM, Tang JM, Zisimopoulos P, King A, Al Assaad M, Teneyck T, Roberts D, Monge J, Demichelis F, Tam W, Ouseph MM, **Sigaras A**, Beltran H, Rennert H, Lindeman N, Song W, Solomon J, Mosquera JM, Kim R, Catalano J, Hassane DC, Sigouros M, Elemento O, Alonso A, Sboner A. EXaCT-2: An augmented and customizable oncology-focused whole exome sequencing platform. ***medRxiv***. 2025:2024.12.05.24318515. doi: 10.1101/2024.12.05.24318515.

[J.30] Anibal J, Gunkel J, Awan S, Huth H, Nguyen H, Le T, Bélisle-Pipon JC, Boyer M, Hazen L; **Bridge2AI Voice Consortium**; Bensoussan Y, Clifton D, Wood B. The doctor will polygraph you now. ***Npj Health Syst.*** 2024;1(1):1. doi: 10.1038/s44401-024-00001-4. Epub 2024 Dec 5. PMID: 39759269; PMCID: PMC11698301.

[J.29] Patel R, Price N, Bahr R, Bedrick S, Bensoussan Y, Bélisle-Pipon JC, Dorr D, Jackson C, Krussel A, Salvi Cruz S, Toghranegar J, Watts S, Zhao R; **Bridge2AI-Voice Consortium**; Powell M. Summary of Keynote Speeches from the 2024 Voice AI Symposium, presented by the Bridge2AI-Voice Consortium. ***Front Digit Health***. 2024 Dec 3;6:1484503. doi: 10.3389/fdgth.2024.1484503. PMID: 39691415.

[J.28] Annabestani M, Sriram S, Caprio A, Janghorbani S, Wong SC, **Sigaras A**, Mosadegh B. High-fidelity pose estimation for real-time extended reality (XR) visualization for cardiac catheterization. ***Sci Rep.*** 2024 Nov 6;14(1):26962. doi: 10.1038/s41598-024-76384-z. PMID: 39505924; PMCID: PMC11542031.

[J.27] Annabestani M, Sriram S, Wong SC, **Sigaras A**, Mosadegh B. Advanced XR-Based 6-DOF Catheter Tracking System for Immersive Cardiac Intervention Training. ***arXiv preprint*** arXiv:2411.02611. 2024 Nov 4.

[J.26] Bahr R, Anibal J, Bedrick S, Bélisle-Pipon JC, Bensoussan Y, Blaylock N, Castermans J, Comito K, Dorr D, Hale G, Jackson C, Krussel A, Kuman K, Komarlu AR, Lerner-Ellis J, Powell M, Ravitsky V, Rameau A, Reavis C, **Sigaras A**, Cruz SS, Vojtech J, Urbano M, Watts S, Zhao R, Toghranegar J; Bridge2AI-Voice Consortium. Workshop summaries from the 2024 voice AI symposium, presented by the Bridge2AI-voice consortium. ***Front Digit Health***. 2024 Oct 30;6:1484818. doi: 10.3389/fdgth.2024.1484818. PMID: 39540145; PMCID: PMC11557516.

[J.25] Awan SN, Bahr R, Watts S, Boyer M, Budinsky R; **Bridge2AI-Voice Consortium**; Bensoussan Y. Evidence-Based Recommendations for Tablet Recordings From the Bridge2AI-Voice Acoustic Experiments. ***J Voice***. 2024 Sep 20:S0892-1997(24)00283-2. doi: 10.1016/j.jvoice.2024.08.029. Epub ahead of print. PMID: 39306498; PMCID: PMC11922786.

[J.24] Rajendran S, Brendel M, Barnes J, Zhan Q, Malmsten JE, Zisimopoulos P, **Sigaras A**, Ofori-Atta K, Meseguer M, Miller KA, Hoffman D, Rosenwaks Z, Elemento O, Zaninovic N, Hajirasouliha I. Automatic ploidy prediction and quality assessment of human blastocysts using time-lapse imaging. ***Nat Commun***. 2024 Sep 5;15(1):7756. doi: 10.1038/s41467-024-51823-7. PMID: 39237547; PMCID: PMC11377764.

[J.23] Evangelista EG, Bélisle-Pipon JC, Naunheim MR, Powell M, Gallois H; **Bridge2AI-Voice Consortium**; Bensoussan Y. Voice as a Biomarker in Health-Tech: Mapping the Evolving Landscape of Voice Biomarkers in the Start-Up World. ***Otolaryngol Head Neck Surg***. 2024 Aug;171(2):340-352. doi: 10.1002/ohn.830. Epub 2024 Jun 1. PMID: 38822764.

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[P.1] A Reiter, **A Sigaras** and P Allen. Surgical Structured Light (SSL) for Real-Time Minimally-Invasive 3D Imaging. ***John Jones Symposium, Columbia University***, 10 May 2013

# Talks, Lectures

## Invited Talks

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|---|---|
| 03/05/2026 | **FutureCare: How AI + XR Are Reimagining Health and Healing**<br>First Thursdays<br>Next Jump NYC HQ |
| 07/17/2025 | **AI & Medical Extended Reality in the era of Precision Surgery**<br>Society of Robotic Surgery (SRS) 2025 Annual Meeting – Strasbourg, France<br>Palais du Congress Convention Center - Marie Curie B |
| 06/09/2024 | **Precision Healthcare - AI and XR interventions**<br>2024 PHIE - Transforming Health Professional Education, Research and Care, São Paulo, Brazil |
| 06/30/2023 | **AI and Medical Extended Reality in the Era of Precision Medicine**<br>36th Annual Congress of the Hellenic Neurosurgical Society & 16th Annual Neurosurgery Nurses Meeting, Athens Greece |
| 10/01/2022 | **AI and XR opportunities in Clinical Education and Care Delivery**<br>Virtual Healthcare in the Mainstream Symposium & Research Forum 2022<br>Weill Cornell Medicine |
| 03/18/2021 | **Metaverse: The Next Frontier in Medicine?**<br>Institute for Computational Biomedicine, Research in Progress Seminar Series<br>Weill Cornell Medicine |
| 06/29/2021 | **XR in Medicine in the era of Precision Medicine**<br>Stanford Healthcare Innovation Lab |
| 05/16/2021 | **XR in Medicine in the era of Precision Medicine**<br>9th International Summer School in Medical Biosciences Research & Management<br>World Hellenic Biomedical Association |
| 01/15/2021 | **XR in Medicine in the era of Precision Medicine**<br>Institute for Computational Biomedicine, Research in Progress Seminar Series<br>Weill Cornell Medicine |
| 07/24/2020 | **XR in Medicine**<br>2nd Biomedical Big Data Symposium<br>New York City College of Technology, City University of New York |
| 02/14/2020 | **AI and Mixed Reality in Precision Medicine**<br>2nd Biomedical Big Data Symposium<br>New York City College of Technology, City University of New York |
| 01/31/2020 | **Mixed Reality in Precision Medicine**<br>NYU Tandon School of Engineering, New York, NY |

10/28/2019 – 10/29/2019 **Digital Technologies for Clinical Care** – Panelist
**Artificial Intelligence for Cancer** – Discussion Leader
Precision Medicine Research Symposium
Cornell Tech and Weill Cornell Medicine, New York, NY

10/25/2019 – 10/26/2019 **XR in Healthcare**
R Lab Master Class 2019

06/05/2019 **XR in Medicine in the era of Precision Medicine**
R Lab Well 2019

06/19/2019 **Big Data Visualization and Spatial Computing in Precision Medicine**
Biomedical Big Data Summer Bootcamp 2019
Weill Cornell Medicine

02/15/2019 **Big Data in Precision Medicine**
1st Biomedical Big Data Symposium
New York City College of Technology, City University of New York

06/20/2018 **Big Data Visualization - Mixed Reality, Virtual Reality and Augmented Reality**
Biomedical Big Data Summer Bootcamp 2018
Weill Cornell Medicine

01/11/2017 **How HoloLens transforms Healthcare**
Microsoft Reactor, Grand Central Tech, NY

11/21/2008 **Introduction to Robotics with MS Robotics Studio**
Microsoft, Athens, Greece

01/29/2008 **Robotics Warehouse Project**
Selected presentation/talk to the Chairman of Microsoft, Bill Gates and Kostas Karamanlis, Prime Minister of Greece at the inaugural opening of the Microsoft Innovation Center in Greece
Microsoft Innovation Center, Greece

12/17/2007 **Introduction to Robotics with MS Robotics Studio and LEGO Mindstorms NXT**
IEEE hosted event
University of Patras, Greece

## Guest Lectures

02/18/2026 **AI and Medical Extended Reality to Care, Discover, and Teach**
CMBP 5005: Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

02/27/2025 **Designing for AI and Medical Extended Reality in the Era of Precision Medicine**
INFO 5358: 3D Interaction Design, Cornell Tech
Host: Prof. Harald Haraldsson

02/05/2025 **AI and Medical Extended Reality in the Era of Precision Medicine**
CMBP 5005: Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

03/06/2024 **AI and Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof. Eugenia Giannopoulou

01/31/2024 **AI and Medical Extended Reality in the Era of Precision Medicine**
Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

11/29/2023 **AI and Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof. Pegah Khosravi

03/01/2023 **AI and Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof: Pegah Khosravi

02/01/2023 **AI and Medical Extended Reality in the Era of Precision Medicine**
Functional Interpretation of High-Throughput Data, Weill Cornell Medicine
Host: Prof: Andrea Sboner

12/05/2022 **AI and Medical Extended Reality in the Era of Precision Medicine**
Topics in Pathophysiology, Deree College, The American College of Greece
Host: Prof: Alexia Victoria Polissidis

02/16/2022 **Medical Extended Reality in the Era of Precision Medicine**
Biomedical Informatics Colloquium, New York City College of Technology, City University of New York
Host: Prof. Eugenia Giannopoulou

11/17/2021 **Medical Extended Reality in the Era of Precision Medicine**
Data Structures and Algorithms for Computational Biology, Weill Cornell Medicine

Host: Prof. Iman Hajirasouliha

10/13/2021 **Medical Extended Reality in the Era of Precision Medicine**
Virtual and Augmented Reality, Cornell Tech
Host: Prof. Harald Haraldsson

12/04/2020 **XR in Medicine in the era of Precision Medicine**
Hostos Community College, City University of New York
Hosts: Prof. Yoel Rodriguez, Prof Anna Ivanova

11/21/2019 **Mixed Reality in Precision Medicine**
Virtual and Augmented Reality, Cornell Tech
Host: Prof. Harald Haraldsson

12/2/2015 **Precision Medicine**
Bioinformatics II, New York City College of Technology, City University of New York
Host: Prof. Eugenia Giannopoulou

11/24/2011 **Intro to Programming in C++ and C#**
Introduction to Programming, University of Piraeus, Greece
Host: Prof. Ioannis-Christos Panagiotopoulos

12/16/2010 **Introduction to Expression Blend**
Human Computer Interaction, University of Piraeus, Greece
Host: Prof. Maria Virvou

12/09/2010 **Introduction to Sketchflow**
E-learning, University of Piraeus, Greece
Host: Prof. Symeon Retalis

## Selected Press and Media Coverage

09/16/2025 **Weill Cornell Medicine Enterprise Innovation News,** *Exploring Human Voice as a Biomarker*
https://innovation.weill.cornell.edu/news-events/exploring-human-voice-biomarker

07/01/2024 **Impact Weill Cornell Medicine Magazine, Summer 2024,** *Features – The Sounds of Science*
http://impact.weill.cornell.edu/summer-2024/features/sounds-science

12/19/2022 **Weill Cornell News**, *Harnessing Artificial Intelligence Technology for IVF Embryo Selection.*
https://news.weill.cornell.edu/news/2022/12/harnessing-artificial-intelligence-technology-for-ivf-embryo-selection

09/13/2022 **NIH News Releases -** *NIH launches Bridge2AI program to expand the use of artificial intelligence in biomedical and behavioral research.*
https://www.nih.gov/news-events/news-releases/nih-launches-bridge2ai-program-expand-use-artificial-intelligence-biomedical-behavioral-research

09/13/2022 **USF Health News**, *USF Health, Weill Cornell Medicine Earn Inaugural Funding in NIH's Newly Launched Bridge2AI Initiative, will Create Artificial Intelligence Platform for Using Voice to Diagnose Disease.*
https://hscweb3.hsc.usf.edu/blog/2022/09/13/usf-health-cornell-earns-inaugural-nih-funding-to-create-artificial-intelligence-platform-for-using-voice-to-diagnose-disease/

09/13/2022 **Weill Cornell News**, *USF Health, Weill Cornell Medicine Earn Inaugural Funding in NIH's Newly Launched Bridge2AI Initiative, will Create Artificial Intelligence Platform for Using Voice to Diagnose Disease.*
https://news.weill.cornell.edu/news/2022/09/usf-health-weill-cornell-medicine-earn-inaugural-funding-in-nih%E2%80%99s-newly-launched

03/07/2022 **Bloomberg***, The Next Life-Saving Innovation in Medicine May Be Developed Only for You.*
https://sponsored.bloomberg.com/article/weill-cornell-medicine/the-next-life-saving-innovation-in-medicine-may-be-developed-only-for-you

08/30/2021 **New York Times***, Experts still aren't sure about corporate meetings taking place in the metaverse.*
https://www.nytimes.com/live/2021/08/30/business/economy-stock-market-news#experts-still-arent-sure-about-corporate-meetings-taking-place-in-the-metaverse

01/30/2021 **NBC News, Peacock - The Overview,** *What Does the Future of Work Look Like?*
https://www.peacocktv.com/watch/playback/vod/GMO_00000000363812_01/b2c4bc2c-dafc-3391-900b-fde591ea42ef

05/28/2020 **Protocol,** *Coronavirus sent us home. Will VR bring us back together?*
https://www.protocol.com/vr-headset-future-of-work

04/06/2020 **Macaulay Honors College News and Events**, *Macaulay Honors College Students to Assist With COVID-19 Effort*
https://macaulay.cuny.edu/news-and-events/macaulay-honors-college-students-to-assist-with-covid-19-effort/

04/01/2020 **New York Times**, *How Are You Feeling? Surveys Aim to Detect Covid-19 Hot Spots Early.*
https://www.nytimes.com/2020/04/01/world/middleeast/coronavirus-detection-surveys.html

07/15/2019 **CNN,** *Would you trust an algorithm to diagnose an illness?*
https://www.cnn.com/2019/07/15/business/artificial-intelligence-healthcare/index.html

04/24/2019 **NYP - Health Matters,** *Behind the Latest Advance in IVF Treatment*
https://healthmatters.nyp.org/behind-the-latest-advance-in-ivf-treatment/

04/11/2019 **Cornell Chronicle,** *Researchers test using AI to optimize IVF embryo selection*
https://news.cornell.edu/stories/2019/04/researchers-test-using-ai-optimize-ivf-embryo-selection

04/05/2019 **Wired,** *AI Could Scan IVF Embryos to Help Make Babies More Quickly*
https://www.wired.com/story/ai-could-scan-ivf-embryos-to-help-make-babies-more-quickly/

04/05/2019 **AI in Healthcare,** *AI optimizes embryo selection for IVF*
https://www.aiin.healthcare/topics/imaging/ai-optimizes-embryo-selection-ivf

04/04/2019 **Wall Street Journal,** *IVF Often Doesn't Work. Could an Algorithm Help?*
https://www.wsj.com/articles/ivf-often-doesnt-work-could-an-algorithm-help-11554386243

04/04/2019 **Weill Cornell News**, *Artificial Intelligence Approach Optimizes Embryo Selection for IVF*
https://news.weill.cornell.edu/news/2019/04/artificial-intelligence-approach-optimizes-embryo-selection-for-ivf

04/04/2019 **Science Daily**, *Artificial intelligence approach optimizes embryo selection for IVF*
https://www.sciencedaily.com/releases/2019/04/190404161003.htm

03/09/2018 **Microsoft Developer Blog**, *Voronoi Selection for Cancer Drug Network Visualization in Mixed Reality*
https://www.microsoft.com/developerblog/2018/03/09/voronoi-selection-cancer-drug-network-visualization-mixed-reality/

10/17/2017 **Channel 9**, *Behind the Scenes: How Weill Cornell Medicine built a chatbot for clinicians to gain fast access to medical data*
https://channel9.msdn.com/Blogs/DevRadio/DR1747

09/13/2017 **Microsoft Windows Blogs**, *Making mixed reality: a conversation with Alexandros Sigaras and Sophia Roshal*

https://blogs.windows.com/windowsexperience/2017/09/13/making-mixed-reality-conversation-alexandros-sigaras-sophia-roshal/#B0hV1JuJsZoIfG2Q.97

06/15/2017 **Microsoft Technical Case Studies**, *Bot gives Weill Cornell clinicians fast access to medical data*
https://microsoft.github.io/techcasestudies/bot%20framework/2017/06/15/WeillCornell.html

06/14/2017 **The Sociable**, *How Mixed Reality is Transforming Collaborative Cancer Research*
https://sociable.co/technology/mixed-reality-cancer/

03/26/2017 **Arirang, Special Documentary** *Smart with Heart*
https://www.youtube.com/watch?v=gHa4ddmdCUU

Winter 2016 **Weill Cornell Medicine Magazine, Volume 15, Number 3,** *Virtual Vision – 3D technology could offer a potent weapon in cancer care*
http://www.weillcornellmedicine-digital.com/weillcornellmedicine/vol_15_no_3?pg=20#pg20

01/27/2016 **ABC7 Eyewitness News**, *New 3D goggles help doctors search for mutations in cancerous tumors*
http://abc7ny.com/health/new-3d-goggles-help-doctors-search-for-mutations-in-cancerous-tumors/1176716/

07/15/2016 **Weill Cornell News**, *Researchers are Using Virtual Reality to Help Treat Cancer*
https://weillcornell.org/news/researchers-are-using-virtual-reality-to-help-treat-cancer

07/13/2016 **CBS New York**, *Seen At 11: Doctors Using Virtual Reality To Treat Cancer Patients*
http://newyork.cbslocal.com/2016/07/13/virtual-reality-cancer-patients/

02/24/2016 **Popular Science**, *Here's How Virtual Reality Could Help Doctors Treat Cancer*
https://www.popsci.com/how-virtual-reality-could-help-doctors-treat-cancer

05/21/2015 **Science Daily**, *Scientists unveil prostate cancer's 'Rosetta Stone'*
https://www.sciencedaily.com/releases/2015/05/150521133732.htm